

Valid Arguments & Fallacies

Name: _____ Date: _____

Patience rule: To claim an argument is valid, show that in every row where all premises are T, the conclusion is T. To claim invalidity, exhibit one row (or one story) where premises are T and conclusion F. Name the fallacy only after that check.

Deduction problems — Arguments

D5.1 DEDUCTION

Consider this argument (define your own letters):

- P1: If I study, then I pass.
- P2: If I pass, then I graduate.
- P3: If I graduate, then I get a job.
- P4: I study.
- Conclusion: I get a job.

1. Symbolize P1–P4 and the conclusion.
2. Derive the conclusion in **numbered steps**, naming each inference (e.g. modus ponens) as you use it. You need at least three steps.
3. Build a truth table for the four atomic letters involved. Verify: in every row where P1–P4 are all T, the conclusion is T. (You may use a 4-row table if you reduce to two letters — explain any reduction.)

SYMBOLS + NUMBERED PROOF + VALIDITY TABLE

D5.2 DEDUCTION

For each argument below: (i) symbolize, (ii) valid or invalid, (iii) if invalid name the fallacy and give a concrete counterexample story, (iv) if valid, give one modus ponens / modus tollens chain.

- A. If the alarm is on, someone opened the door. The alarm is on. So someone opened the door.
- B. If the alarm is on, someone opened the door. Someone opened the door. So the alarm is on.
- C. If the alarm is on, someone opened the door. The alarm is not on. So nobody opened the door.
- D. If the alarm is on, someone opened the door. Nobody opened the door. So the alarm is not on.

Process all four — they look similar on purpose.

FOUR STRUCTURED ANALYSES (I)–(IV)

D5.3 DEDUCTION

Negation + De Morgan drill (show every rewrite step):

Original English: “It is false that the shop is open and (the sale is on or the manager is here).”

1. Let o = shop open, s = sale on, m = manager here. Write the original as $\neg(o \wedge (s \vee m))$.
2. Apply De Morgan’s laws twice until you have a formula using only $\neg$, $\wedge$, $\vee$ with **no** parentheses around a negated AND/OR group.
3. Translate the final form back into plain English.
4. Pick one assignment of T/F to o , s , m and evaluate **both** the original and the final form step by step to confirm they match.

REWRITES + ENGLISH + EVALUATION CHECK

D5.4 DEDUCTION

Longer argument — test validity carefully:

- P1: Everyone who plays chess enjoys strategy games.
- P2: No one who enjoys strategy games hates puzzles.
- P3: Maya hates puzzles.
- Conclusion: Maya does not play chess.

Use letters: c = Maya plays chess, s = Maya enjoys strategy games, h = Maya hates puzzles. Translate P1–P3 and the conclusion as best you can with $\rightarrow$ and $\neg$ (P2 is a “no one” statement — rewrite as “If someone enjoys strategy games, they do not hate puzzles” for Maya).

1. Write the symbolized argument.
2. Attempt a step-by-step derivation of the conclusion from the premises. If you get stuck, say where.
3. Build a 3-variable truth table (8 rows) for the conjunction of premises $\rightarrow$ conclusion. Is the argument valid?
4. If valid, summarize the proof in three sentences for a non-expert. If invalid, give a counterexample row.

SYMBOLIZATION + DERIVATION ATTEMPT + 8-ROW TEST + SUMMARY